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Effects of trust and perceived self-efficacy on aided judgement: an experiment on spelling and grammar checking

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ABSTRACT

We report an experimental study in which participants make a series of linguistic judgements with help from a fictional text-editing aid. The study illuminates how acceptance of a software aid's advice might be affected by, and in turn affects, users' trust and confidence.

We measure the propensity to accept automated advice by adapting Signal Detection Theory (SDT), and particularly the construct of bias. Our study demonstrates a significant difference in bias between groups who interact with aids that perform at different levels of reliability. Participants' pre-task trust in similar systems also affected bias towards the aid, but we found no evidence for such an effect of prior perceived self-efficacy.

Users show an awareness of their own performance, demonstrated by higher self-reported confidence in correct responses than in incorrect ones. Further, participants using a more reliable aid are more confident when accepting than when rejecting the aid's advice. Overall, participants were overconfident in their judgements.

Our research demonstrates opportunities of using an experimental paradigm, analysed with SDT, to study aspects of performance, trust, and confidence in knowledge-based cognitive tasks aided by an artificially intelligent aid.

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1. Introduction

A common feature of contemporary Human–Computer Interaction (HCI) is the provision of artificially intelligent (AI) advice that users can consider before making a decision. An extremely familiar version of this situation is the use of spelling and grammar checkers in word processing software. HCI research in this domain is often concerned with maximising utility of the AI (acceptance of advice, whether good or bad) and improving judgement of the advice (Schemmer et al. 2023; Vorm and Combs 2022). In this article, we look in particular at how the underlying mechanism of deciding to accept an aid's spelling or grammar advice might be affected by, and in turn affect, users' trust and confidence.

The more general aim of the reported experiment is to examine how human decision makers' decisions under uncertainty and their confidence in these decisions might be affected by an automated system's advice, especially when that advice itself is somewhat uncertain or imperfect. We use Signal Detection Theory (SDT; [Macmillan and Creelman 2005]) to analyse performance and confidence in a cognitive linguistic judgement task, and show how the construct of *bias*

can be appropriated to measure users' willingness to accept or reject an aid's advice.

For this research we have developed an experimental paradigm, based on the familiar concept of suggestions for spelling and grammar alternatives in word processors. We measure how participants' performance when working with our system is influenced by their trust in similar systems and their judgement of their own efficacy prior to the task. We also test the confidence participants have in their decisions in conjunction with the aid's advice. We regard user confidence, and in particular the calibration of user confidence with performance, to be another key metric for the usability of advice systems; our paradigm allows this idea to be explored. We measure self-reported confidence in the domain of interaction in relation to performance in three distinct ways: prior to a task (as *perceived self-efficacy*), with respect to each single decision during the task, and post-task integrated over a complete series of decisions.

To anticipate our most general findings, we observe that users acknowledge the aid and gain an appreciation of its level of performance, as is demonstrated by a

higher propensity to follow a better performing aid's advice. Although participants are overconfident in the sense that their mean trial confidence as well as their post-task frequency estimation of correct responses are higher than warranted by their performance, their ability to value their own judgements is evidenced by a higher level of confidence in correct decisions than in incorrect ones.

2. Background

2.1. Automated assistance in advisory interactions

This article is concerned with *advisory interactions* (Woods, Johannesen, and Potter 1991), where a human is the ultimate decision maker, and an AI system takes the role of a consultant or advisor. Automated decision aids can perform two discretely different functions, often in tandem: to *alert*, and to *advise* (or *recommend*). The purpose of the alert function is to indicate a situational change that may require action on behalf of the user, and the advisory function serves to make suggestions on types of actions or alternatives (Parasuraman and Manzey 2010). An example of an alert is a red wavy underline under a potentially misspelt word in MS Word or a similar text editor, and an example of advice is the list of alternative spellings the system provides when the underlined word is clicked on or hovered over. The aid's *reliability* is the measure of its performance, i.e. the percentage of its advice being correct.

2.2. Trust in automated decision aids

Trust is the expectation that some other agent (in our context: the automated aid) will demonstrate technically competent performance, and that they will act in good faith, i.e. that they will let common interest prevail above their own interest (Barber 1983; Muir 1987). Trust is widely recognised as an important factor in the interaction between humans and systems (see e.g. Bisantz and Seong 2001; Chavaillaz, Wastell, and Sauer 2016; Dzindolet et al. 2003; Hoff and Bashir 2015; Kohn et al. 2021; Lee and Moray 1992; Lee and Moray 1994; Lee and See 2004; Madhavan and Wiegmann 2007; Moray et al. 1994; Moray and Inagaki 1999; Muir 1987): if a decision aid is not trusted by a user, they may reject its advice, or use alternative means that are potentially less time and energy efficient.

Overreliance on an aid, or *misuse*, may lead to less than ideal performance, and underreliance, or *disuse*, is arguably equally problematic (Parasuraman and

Riley 1997). Parasuraman and Manzey (2010) observe that overreliance, where users' critical analysis is replaced with blind trust in the automated system, may lead to two types of error: *error of omission*, and *commission error*. An error of omission is a situation where a user fails to act because they have not received an alert from the aid, e.g. a text editor fails to underline a misspelt word and the user ignores the error. This may even be the case if the error is obvious, but the user decides the system 'must know best'. A commission error denotes the inverse, e.g. a text editor underlines a word that is obviously correct, but the user still acts on the system's alarm and accepts its recommendation.

The twin possibilities of under and overuse of an automated decision aid have been the main focus of a body of research connected with the Theory of Technology Dominance (e.g. (Cabitza et al. 2023; Hampton 2005; Sutton, Arnold, and Holt 2022)). This work focuses on measuring reliance on an aid by comparing initial unaided judgment with 'final' judgments in which advice is considered. Our work is distinct from this tradition, in that it measures propensity to accept advice without identifying unaided judgements, and uses Signal Detection Theory to test interaction with the aid.

It is widely acknowledged that trust in automation is likely to combine an initial, transferred level of trust, with a particular trust that develops during use of the system. Initial trust is presumably transferred from experience with similar or perceived-to-be-similar systems (Lee and See 2004), and might also be affected by individual factors such as users' beliefs or implicit attitudes towards automation (Hoff and Bashir 2015; Merritt et al. 2013). Trust developed during use will be influenced by factors such as user personality (Merritt and Ilgen 2008; Merritt et al. 2015), and of course the aid's performance, or more precisely, users' perception of its performance (De Vries, van den Berg, and Midden 2015; Hutton and Klein 1999; Li, Hess, and Valacich 2008; Moray, Lootsteen, and Pajak 1986), which implies that users can construct a sense of an aid's reliability (Hutchinson et al. 2022). Trust is a dynamic factor (Yang, Schemanske, and Searle 2023), and perception of an aid's performance is partly guided by the errors users see a system make (De Vries, Midden, and Bouwhuis 2003; Dzindolet et al. 2003; Johnson et al. 2004; Rice and McCarley 2011; Wiegmann, Rich, and Zhang 2001).

2.3. Metacognition: perceived self-efficacy and confidence

Metacognition describes *cognition about cognition* (Cox 2005), or a person's capacity to recognise their own

successful cognitive processing (Fleming and Lau 2014) and effectively evaluate the accuracy of one's own inferences (Garcia-Retamero, Cokely, and Hoffrage 2015). Schraw and Moshman (1995) observe a distinction between (a) metacognitive knowledge, and (b) metacognitive control processes. In our study, we have researched both categories in the form of (a) participants' beliefs about their abilities prior to an experimental task (perceived self-efficacy), and (b) their confidence during and after the task.

If trust in automation is likely to increase the use of the automation, perhaps 'trust' in one's own unaided competence (*perceived self-efficacy*) or performance (*confidence*) is likely to diminish any such reliance. We use the term *perceived self-efficacy*, which we explain in more detail in section 2.3.1, when we talk about a person's personal pre-task beliefs about their own ability in relation to the task, and the term *confidence*, which we explain in section 2.3.2, when we discuss people's personal beliefs about the accuracy of their judgements (Allwood and Montgomery 1987; Stankov, Kleitman, and Jackson 2015). To avoid confusion, in this article we avoid the common phrase *self-confidence*, which can be ambiguously used to describe perceived self-efficacy as well as confidence. For internal consistency, we follow this logic also when referring to literature that discusses the same concepts but uses these terms differently or interchangeably.

Although the interplay between users' trust and perceived self-efficacy is widely acknowledged as an important factor in predicting operators' task allocation strategies in joint human-automation systems in general (e.g. Lee and Moray [1994]; Moray et al. [1994]; Wiczorek and Meyer [2019]), there is no universal agreement on the exact nature of this relationship, and it is not always fully clear if perceived self-efficacy or confidence is measured.

It must be noted that trust and perceived self-efficacy are not the only moderators in the discretionary use of decision aids. Factors like workload can also play a key role in disuse and misuse (Parasuraman and Riley 1997; Sarter, Woods, and Billings 1997), for example because they might lead to a form of effort reduction (Davis and Tuttle 2013). However, these factors are outside the remit of the current study.

2.3.1. Perceived self-efficacy – personal beliefs about ability

Bandura (1997) coined the phrase *perceived self-efficacy* to describe 'a function of domain-specific beliefs about personal capacities' (Carroll and Reese 2003), or, in other words, he defines it as a person's own estimation of their ability to 'get a job done satisfactorily'.

While (*self-*)*efficacy* refers to a person's actual ability, *perceived self-efficacy* refers to the judgement that a

person makes of their own *specific* proficiency level, so it does not denote their actual skill or knowledge level, or their general global level. As a result, there is no single measure of perceived self-efficacy (Bandura 2006). Actual self-efficacy can be built over the years, whereas perceived self-efficacy is much more dynamic and can even change instantly, for example when a person gets stuck in a process, or successfully completes or fails a task (*enactive mastery experiences*, see e.g. Bandura (1997) and Bandura (2006)). Although perceived self-efficacy may serve as a proxy for ability in some cases, it cannot be treated as such by default. However, Bandura (1984) claims that perceived self-efficacy is a reliable predictor not only of people's causal attribution of successes and errors, but also of actual performance.

2.3.2. Confidence judgements – evaluating decisions

Confidence, in our sense, is a person's post-hoc degree of belief about the level of their judgement or decision, or their overall post-task performance. In the literature, two sub-classes of confidence are identified, (a) single-event judgements of confidence, usually measured as probability of correctness after each trial in a task, and (b) judgements of overall confidence, expressed as frequencies of correct responses, measured after completion of a task consisting of sequence of trials (Gigerenzer 1991, 1994; Gigerenzer, Hoffrage, and Kleinbölting 1991).

2.3.3. Overconfidence and confidence calibration

Following on from the breakdown of confidence evaluations into single event probability and long-term frequency measures, we identify a similar division of *overconfidence* measures in the literature: (1) pre-task, or forecasting, reports that are essentially measures of *overassessment of perceived self-efficacy* (see e.g. Dunning [2011]), and (2) post-hoc measures of overconfidence that represent *overestimations of performance* (see e.g. Moore and Healy [2008]). We subdivide the latter into (2a) post trial, or *online*, and (2b) post-task, or *post experiment*, appraisals, which do not affect the interaction with the aid.

Online confidence judgements are spontaneous and subjective, and can thus never be characterised as 'true' or 'false', although the correctness of the response to which the probability is assigned might be checked later (Lichtenstein, Fischhoff, and Phillips 1977b). Despite Lichtenstein et al.'s assertion that individual probabilities themselves cannot be invalidated, they suggest that a participant's level of calibration can be checked by comparing the means of their confidence ratings during a task with actual performance measures (i.e. percentage of correct responses). A judge (participant), they argue, is well-calibrated if 'over the long run, for all propositions assigned a

given probability, the proportion that is true is equal to the probability assigned’.

Where it is clear that post-task judgements of confidence cannot, by definition, affect decision-making and interaction with an aid, whether this is true for online confidence evaluations is much less clear. Double and Birney (2019) suggest that there is a risk of reactivity when confidence is measured, or, in other words, that measuring confidence might facilitate (i.e. positively affect) or impair (i.e. negatively affect) participants’ judgements. Testing whether, or to what extent, this type of reactivity might be a confounding factor in decision-making in our experiment was outside the scope of the current research, as was in-depth analysis of other potential methodological artifacts such as variants of the observer effect.

2.4. Research questions

In summary, we formulated two research questions around the simple model of an aided linguistic decision-making process based on a balance of perceived self-efficacy and trust. Firstly, to what extent do perceived self-efficacy and trust in similar systems affect interaction with the aid, and in particular, the propensity to accept the aid’s advice? And secondly, to what extent does the aid’s reliability affect users’ decision-making, and the confidence they have in their decisions?

There is a methodological challenge in measuring propensity to accept an aid’s advice on tasks that users can attempt unaided, but for which unaided performance is not available or measured. If unaided performance is above chance, correct advice will appear to be accepted more often. To address this, we utilise Signal Detection Theory to separate discrimination from bias, and to align bias with the tendency to respond in line with the aid’s advice, independent of advice correctness.

3. Experimental study

3.1. Method

We designed an online experimental study in three parts. In *Pre-task* it is established who participants are, and some of the relevant beliefs they hold are identified. In the second part, *Task*, participants execute a series of linguistic judgement tasks. Lastly, in *Post-task*, participants judgements of their own and the system’s performance during the task are measured.

The study used an off-line Wizard of Oz method, in which pre-selected suggestions are described as being made by an automated system. Ethical approval for this minor deception was gained through the University

of Bath Department of Computer Science ethics process (EIRA1), and participants were fully debriefed.

3.2. Participants

120 participants (60 Male/ 60 Female; age 19–71, $M = 38.95$, $SD = 13.03$; GroupGood (GGood), $M = 30$ / $F = 29$, age $M = 40.46$, $SD = 12.86$; GroupBad (GBad), $M = 30$ / $F = 31$, age $M = 37.44$, $SD = 13.20$; a definition of the experimental conditions for GGood and GBad is given in section 3.3.2) were recruited by listing the experiment on the online survey participant recruitment platform Prolific.ac. Sample size was determined with the software G*Power to give a power of at least 0.8 for the simple Human Factors hypothesis that a more accurate decision aid would be beneficial to participants’ overall judgement performance if this were tested using a *t*-test for a medium size effect.

The Prolific.ac listing specified the general topic and the task of the study, the estimated completion time (18 minutes, automatically estimated by Qualtrics) and the maximum completion time (25 minutes), and the reward (£1.80 payment upon completion of the survey, which was, following Prolific’s terms, only paid if the survey was taken in full).

Participants were pre-screened on location (registered as United Kingdom resident), nationality (registered as United Kingdom citizen) and first language (English) to ensure their responses would be credible with regards to command of British English. This pre-screening was arguably not perfect as it would potentially allow speakers of other variants of English to be part of the sample, but it was as precise as possible within Prolific at the time. Additionally, participants were also pre-screened with regards to participation in previous studies, and those who participated in the pilot study (see section 3.5) were excluded.

Participants’ education level was generally high (vocational training/ apprenticeship/ higher education degree), and they all had experience using word processing software such as Microsoft Word or Apple Pages. Should a participant report having no prior experience with word processing software, on which the fictional aid Stylus’ behaviour was modelled, their results would not be credible, and their data would be ignored. We did not identify such a case in this study.

3.3. Experimental design

3.3.1. Pre-task

The pre-task questions aimed to establish some of the relevant background characteristics of participants of our studies in terms of:

- (a) identifying their sociodemographic profile (age, gender, and educational level).
- (b) gauging their experience with word processing software (e.g. Microsoft Word and Apache Open Office) to ensure their task responses are credible.
- (c) measuring their self-reported levels of trust in spelling and grammar suggestions in word processing software packages or internet browsers in general, to test our hypotheses for correlations with their acceptance of advice.
- (d) assessing their perceived level of self-efficacy in spelling and grammar, to test our hypotheses for how this interacts with trust of automation to affect acceptance.

Where we asked participants to rate their own and systems' attributes, we used sliders with 0% – 100% scales. Each slider had a pointer set in the middle (50%) as the default, this had to be manipulated before participants could proceed to the next item. If the default was the desired position, participants could move the pointer, then move it back to proceed.

Although several attempts at standardisation have been made in recent years, there are currently no standardised methods for measurements of initial trust in automated systems and trust developed during use (Kohn et al. 2021). Our pre-task trust measurement method was based on common examples of reliable measures from the literature, which all use similar questions, and typically 'not at all agree – completely agree'-type Likert-scales (see e.g. Lichtenstein et al. [1977a], Muir [1987], Muir [1989], Gigerenzer et al. [1991], and Lee and Moray [1992]). The wording of the items was slightly adapted to fit with the domain, and we used 0–100 scales instead of Likert-scales to achieve consistency with our other measures, such as that of trial confidence. We measured participants' pre-task trust with the items 'When thinking of the trustworthiness of spelling suggestions in word processing software packages or internet browsers in general, I would class them as [0; Not very trustworthy at all] – [100; Very trustworthy]' and 'When thinking of the trustworthiness of grammar suggestions in word processing software packages or internet browsers in general, I would class them as [0; Not very trustworthy at all] – [100; Very trustworthy]'. Internal reliability was tested (Cronbach α (unstandardised) from around .70 was assumed to be acceptable (Field 2013) and the mean score was used to compute participants' average level of prior trust. Prior trust was not significantly different in GGood ($\alpha = .88$, $M = 76.42$, $SD = 17.07$) and GBad ($\alpha = .93$, $M = 75.60$, $SD = 16.56$): $t(118) = 0.281$, $p = .779$, $d = 0.051$.

There is no single method to measure perceived self-efficacy, nor a single metric that describes it (Bandura 2006). Participants' pre-task judgement of their own linguistic capability was measured with the items 'When thinking of how good I am at English grammar, I would class myself as [0; Not very good at all] – [100; Very good]' and 'When thinking of how good I am at English spelling, I would class myself as [0; Not very good at all] – [100; Very good]'. Internal reliability was tested following the same method as for prior trust, and the mean of the two scores was used to measure participants' overall level of prior perceived linguistic self-efficacy. Perceived self-efficacy was not significantly different in GGood ($\alpha = .95$, $M = 73.18$, $SD = 14.83$) and GBad ($\alpha = .95$, $M = 73.24$, $SD = 14.25$), as in $t(118) = -0.02$, $p = .981$, $d = -0.004$.

3.3.2. Task

The experimental task had a between-subjects design, for which participants were randomly split into two almost equally sized groups (GGood 59 vs. GBad 61 participants). In both groups, each participant was presented with thirty consecutive trials. In each trial, participants were shown two sentences, of which they had to indicate which one they thought was 'better'. Of the first alternative (labelled 'Original sentence' in each trial) they were told in an introductory paragraph that it was written by a human writer, of the second that it was 'a suggestion for improvement from an automated editing tool called Stylus' ('Stylus suggestion'). No performance feedback was provided to participants during the experiment. An example of a single trial interface is shown in Figure 1.

GGood encountered a version of the experiment in which Stylus correctly suggested an alternative for twenty sentences with errors ('signal trials'), and incorrectly suggested alternatives for ten correct sentences ('noise trials'). GBad received ten correct suggestions and twenty incorrect ones (i.e. a system that performs very poorly). For GBad, ten GGood items were inverted (i.e. the 'Stylus suggestions' for the first group were now presented as the 'Original sentences').

The trials were grouped into five randomly ordered blocks, which were each internally randomised as well. This double form of randomisation served to minimise the chance participants would encounter too many items from the same category (e.g. apostrophe errors, or run-on sentences) at one time during the experiment, thus minimising the risk of item difficulty confounding results.

3.3.2.1. Response confidence. During the experimental task participants were asked after each trial to rate their

Survey Completion 0% 100%

Original sentence: The dentist told her to lay down and relax, which was easier said than done facing the prospect of a root canal treatment.

Stylus suggestion: The dentist told her to lie down and relax, which was easier said than done facing the prospect of a root canal treatment.

Which of the above do you think is better:

Original sentence

Stylus suggestion

How confident are you in your answer?

50% (I guessed) 100% (I'm certain I got it right)

50 55 60 65 70 75 80 85 90 95 100

→

Figure 1. Trial interface example screenshot.

confidence in their response by moving a slider on a 50% ('I guessed') – 100% ('I'm certain') scale. Sliders had the pointer set in the middle (75%) as a default, this had to be manipulated before participants could proceed to the next trial. If the default was the desired position, they could move the pointer and return it to 75% to proceed. This method of measuring trial confidence is commonly used in Cognitive Psychology research (e.g. Gigerenzer et al. [1991] and Gigerenzer et al. [2008]). Values below 50% would not make sense, as participants should in such case select the alternative option.

3.3.2.2. Materials. The design of the decision tasks required examples of correct and incorrect original sentences and rewrites at different levels of writing style and difficulty. We did not have any background data about the kinds of grammatical errors that are most common in our participant sample, nor much prior intuition about how competent the participant pool would be at the general task. Therefore, inspiration for the sentences used in the experiment was drawn from common errors observed on the RetroRides internet forum (2018b), readers' comments on the Guardian newspaper website (2018a), and from the Collins Improve your skills series (King 2009a, 2009b, 2009c). Stylus combines the behaviour of a set of real-world style checkers integrated in word processors or internet browsers (e.g. Word, Grammarly, Hemingway, Grammar, Language Tool, Slick Write, Whitesmoke and Espresso) and suggestions from the Collins books. We freely mixed the suggestions from different sources, so as to gather a wide sample of

particular cases. Errors used in the trials included among others punctuation, spelling, homophone (similar sounding words with distinctly different meanings) and tautology and pleonasm (redundant word) errors.

3.3.3. Post-task

Post-task, participants were asked to estimate their own performance as a measurement of their confidence across the entire experiment (Gigerenzer 1994; Gigerenzer, Hoffrage, and Kleinbölting 2008). They were first asked to estimate how often they chose the Stylus suggestion over the Original sentence ('You have just rated thirty Stylus suggestions for rewrites of original sentences, how often do you estimate you chose the Stylus suggestion over the original sentence? [0] – [30]'). They were then asked to rate their own level of performance ('Not all Stylus suggestions were correct; how often do you estimate you chose the correct answer (either the original sentence or the Stylus suggestion)? [0] – [30]'). This method is commonly used to demonstrate the *overconfidence effect* and how it can 'disappear', by comparing the mean of single-event confidence probabilities and post-task frequency estimations (Gigerenzer 1994; Gigerenzer, Hoffrage, and Kleinbölting 2008). In our analyses, the latter are converted into percentages for the sake of comparison.

Participants' post-task trust in the aid was measured with the items 'When thinking of the trustworthiness of Stylus' performance during this experiment, I would class it as [0; Not very trustworthy at all] – [100; Very trustworthy]' and 'When thinking of the consistency

of Stylus' performance during this experiment, I would class it as [0; Not very consistent at all] – [100; Very consistent]' (GGood: $\alpha = .83$; GBad: $\alpha = .68$).

Lastly, participants were asked 'How plausible do you find it that the Stylus suggestions were created by an automated system? [0] – [100]'; we used this as an indication of their level of engagement with the task. Again, 0% – 100% sliders were used, with the pointer set at 50% as the default. It was reasonable to expect that participants in GGood found it more plausible that the 'Stylus suggestions' came from an automated system than those in GBad, because performance of the aid in the first group is much closer to that of a real-world system than that in the latter. Although the plausibility rating was significantly higher in GGood ($M = 70.95$, $SD = 15.66$) than in GBad ($M = 65.46$, $SD = 21.47$), as in $t(119) = 39.38$, $p < .001$, $d = 3.595$, the difference was smaller than expected.

The same sliders were used as for the pre-task items, and internal reliability of aggregated factors was checked following the same method.

3.4. Hypotheses

The main hypotheses for this experiment derive from the theory discussed in section 2, that use of an automated aid will increase if prior trust in the aid is high, and decrease if the user's prior perceived self-efficacy is high (*H2a, b*: tests of these hypotheses rely on an SDT-based analysis of bias, as will be discussed in more detail in section 4.1). The secondary hypotheses relate to the central Human Factors proposition that a better aid will improve participants' performance (*H1a, b*) and to various aspects of confidence (*H3*) and trust (*H4*). Each hypothesis is tested in section 5 once all required data have been reported.

- H1** Compared to GBad participants, participants in GGood will during the task
- (a) perform better in terms of percentage correct.

This hypothesis concerns the very idea that an automated aid positively affects participants' performance.

- (b) have a stronger bias towards following Stylus' advice, measured using Signal Detection Theory.

This hypothesis concerns the suggestion that an automated aid's performance affects participants' bias towards following the aid. If confirmed, this shows that a more reliable aid might positively affect participants' acceptance of its advice.

- H2** Independent of the correctness of its advice, participants will have a higher propensity to accept the aid's suggestions if

- (a) their prior trust in automated writing style checkers is high.
- (b) their perceived self-efficacy in the domain of writing is low.

- H3** Participants in both groups will be overconfident, especially when confidence is measured trial-by-trial; i.e. their estimates of percentage correct will be higher than actual achieved percentage correct.

- H4** Compared to GBad participants, participants in GGood will develop more trust in Stylus during the experiment, as measured post task.

As well as testing these main hypotheses, additional statistical analyses will be undertaken to investigate the relationship between variables and explore the viability and usefulness of our experimental paradigm.

3.5. Apparatus

The study was conducted fully online, with the survey being created on, and hosted by the survey platform Qualtrics.com, and participants recruited through the recruitment website Prolific.ac.

Before the experiment proper, we ran a pilot study to test item difficulty to ascertain whether the reliability threshold of 0.7 (Wickens and Dixon 2007) was reached. The pilot was deliberately underpowered (62 participants) for reasons of cost, hence the results are not reported in this article. Items that received 100% correct responses were replaced with intuitively more difficult ones. The pilot experiment also allowed us to confirm that the wording and response scales of all questions were understood by participants, and led to well-distributed means with high internal reliability where appropriate.

3.6. Procedure

Participants conducted the experiment remotely on their own device. After accepting participation in the experiment on the Prolific platform, they were automatically assigned to group GGood or GBad by Prolific according to the order in which they accepted participation in the experiment.

To ensure participants completed the study in one sitting, a maximum duration of 25 minutes was set; the actual average completion time was 18.37 minutes ($SD = 6.18$; GGood: $M = 18.0$ ($SD = 5.13$); GBad: $M = 18.37$ ($SD = 6.18$)). For unknown reasons, Prolific allows participants to overrun this limit, and in the case of this

experiment, 14 participants took more time to complete it. Because their prior trust and prior perceived self-efficacy, as well as performance and confidence results were not statistically significantly different from the mean, their results have been kept.

4. Analysis strategy and reporting of results

Independent samples *t*-tests were used to compare conditions between groups, paired samples *t*-tests were used to compare conditions within groups, repeated measures ANOVAs were used to test interaction effects, and Pearson's *r* was used to examine correlations.

Assumption tests were carried out for all analyses. For *t*-tests these were normality (normal distribution of dependent variable; Shapiro–Wilk) and equality of variance (equal variance in all groups; Levene) tests. For ANOVAs they were Mauchly's sphericity tests (equal variances of the differences between all combinations of related groups; with sphericity corrections (none, Greenhouse-Geisser, Huynh-Feldt)), and homogeneity (Levene) tests. Assumption test results are not reported unless they were violated, in which case this is noted; this happened in only a single test, see section 5.2.

4.1. Signal Detection Theory to analyse task data

We used a version of SDT to analyse certain aspects of performance and confidence data.

The first item in each trial was labelled 'Original sentence' and the second item 'Stylus suggestion', and participants had to indicate which one was better (because an error or infelicity was present in the other). Performance on such a trial, in analogy to SDT trials, combines the participant's unaided sensitivity to correctness of sentences with their willingness to accept the advice of an automated system. For example, if the aid's suggestions are correct for 80% of the trials, then a participant could achieve 80% correct by simply accepting each suggestion, even if they had no knowledge of English grammar and spelling.

The logic of SDT analysis can be applied to this situation, to separate sensitivity (ability to identify correct sentences) from bias (propensity to accept automated advice); our adaptation of the SDT matrix is shown in Table 1.

Note that possible categories in our model depend on the type of trial: Original sentence is correct and Stylus suggestion is incorrect, or vice versa. Essentially a trial in which the Stylus suggestion is correct is treated as a 'signal', when the suggestion is incorrect this is treated as 'noise', but these terms have no meaning other than to reflect the mapping to SDT. Possible response rates

Table 1. Signal detection matrix.

	Stylus suggestion correct	Original sentence correct
Stylus suggestion selected by participant	Hit Correctly accepted suggestion	False Alarm Incorrectly accepted suggestion
Original sentence selected by participant	Miss Incorrectly rejected suggestion	Correct Rejection Correctly rejected suggestion

always balance another cell of the same pair (Hit rate (H) and Miss rate (M), or Correct Rejection rate (CR) and False Alarm rate (FA)) depending on this type. SDT trials are either signal or noise: signal trials will result in Hits or Misses, noise trials in False Alarms or Correct Rejections. Misses equal errors of omission, whereas False Alarms are commission errors (Johnson et al. 2004; Parasuraman and Manzey 2010), both of which were introduced in section 2.2.

And finally, to complete the analogy with SDT, from the above matrix we can see that a participant's sensitivity can be computed by subtracting FA from H, and their willingness to accept advice, or bias, can be computed by adding FA to H. Sensitivity (d') in this analysis is performance independently of being aided, and consequently has a minor role in our studies, but it allows us to query the validity of self-efficacy judgements. Bias (criterion, c), however, as propensity to accept advice, is an essential measure for our studies, and arguably a major novel contribution of this research.

Following SDT, the formulae used to compute these measures were as follows:

$$\text{Hit rate: } H = N_H / (N_H + N_M)$$

$$\text{False Alarm rate: } FA = N_{FA} / (N_{FA} + N_{CR})$$

$$\text{Sensitivity: } d' = z_H - z_{FA}$$

$$\text{Bias: } c = z_H + z_{FA}$$

d' and c cannot be computed if $H = 0$, or $FA = 0$. To solve this issue (relevant for 2 participants in our study), the common method of subtracting 0.5 from $H = 0$ or $FA = 0$ was used (Macmillan and Creelman 2005).

If $H = FA$, $d' = 0$, i.e. pure chance, which indicates participants cannot discriminate between good and bad Stylus suggestions and their responses are likely the effect of guessing (Bartlett and McCarley [2017] exclude participants with d' scores below 0.5, because they assume not meeting that lower limit suggests participants have failed to understand or follow the instructions). If $H \geq FA$, d' will be ≥ 0 . The higher a participant's absolute d' -score is, the more sensitive they are to discrimination between the correct answer and the incorrect one; theoretically the highest possible d' -score is infinite, but when above corrections are applied to $H = 0$ and $FA = 0$, 6.93 is the upper limit. However, when $H = 0.99$ and $FA = 0.1$, the effective

upper bound is 4.65. Typical d' -scores are values up to around 2 (Macmillan and Creelman 2005).

If $FA = M$, $c = 0$. If $FA > M$, c will be positive and, in our model, indicate bias towards following the aid, and if $FA < M$, c will be negative, which is indicative of bias towards the Original sentence. Note that the range for c is equal to that of d' , but that the centre is 0, which indicates an absence of bias. When both H and FA are high, c will be positive ($H = 0.99$ and $FA = 0.99$ results in $c = 2.33$), and, conversely, when H and FA are both low, c will be negative ($H = 0.01$ and $FA = 0.01$ results in $c = -2.33$) (Macmillan and Creelman 2005).

4.1.1. Noise in results

The distribution between the proportions of signal and noise trials, or *likelihood ratio*, in our experiment is asymmetric as dictated by the reliability of the aid in each group. This is normally not ideal because it inherently introduces noise in the results (Macmillan and Creelman 2005), but it is an essential part of our method because testing bias towards the aid, i.e. participants' propensity to ignoring or accepting its advice, is our main objective. The low number of trials is another inherent potential source of noise, but we believe this is an acceptable compromise because it doesn't bias the results in one way or another. A low number of trials just lowers power, and we accept this compromise because increasing the number of trials would result in the introduction of yet different sources of noise, e.g. because performance reduces due to fatigue-induced inattentiveness (see e.g. Macmillan and Creelman [2005]).

4.1.2. Artificially poor reliability of the aid

Perhaps an oddity in our experiment, at least from a Human Factors point of view, is the poor performance of the aids the groups encounter. In both groups, the aid performs at a much lower level of reliability than would be acceptable for any real-world system (GGood 0.67; GBad 0.33). The reason for this is two-fold. Firstly, an exaggeration of the aid's underperformance is required to assure a balance between correct and incorrect advice in the experimental conditions. If the aid's reliability would be above the reliability threshold of 0.7 (Wickens and Dixon 2007), there would not be enough data in some of the conditions to be meaningfully analysed, which would compromise the usefulness of our design. Secondly, the difference in reliability of the aid in both groups was a compromise between the requirement of reliability being comparable between the groups, and it being far enough apart for the difference to be meaningful. These choices allow us to test the central idea of our SDT-

based methodology. It should be noted that poor reliability of an aid will negatively affect the user's trust (De Vries, Midden, and Bouwhuis 2003; Dzindolet et al. 2003; Johnson et al. 2004; Kohn et al. 2021; Rice and McCarley 2011; Wiegmann, Rich, and Zhang 2001; Wojton et al. 2020), which should be factored in when interpreting trust results in any experiment on imperfect automated aids.

4.2. Analysing confidence

Participants' confidence ratings were analysed using a mixed repeated measures ANOVA. The within-subjects variables that were tested were Correctness of the response (i.e. 'Correct' vs. 'Incorrect') and Type of response (i.e. 'Stylus' vs. 'Original'). This reconfigures the SDT matrix to allow correctness of response to be a main effect. The between-subjects factor was Group (i.e. GGood vs. GBad). Participants' average confidence in each cell of the design was entered into the ANOVA.

As all participants completed all trials, missing data in confidence cells are not the result of missing responses, but rather the result of high performance and lack of Misses and/ or False Alarms. Therefore, missing values are substituted by the mean of participants' confidence cells that are present. For example: if there was no data in an 'FA confidence' cell, the mean of a participant's 'H confidence', 'M confidence' and 'CR confidence' was used instead. Although this method has its drawbacks, such as the potential to introduce bias, it is commonly used (Donders et al. 2006).

5. Results

5.1. Performance

H1a Compared to GBad participants, participants in GGood will during the task perform better in terms of percentage correct.

The average percentage correct was not significantly higher in GGood ($M = 64.97$, $SD = 16.23$), than in GBad ($M = 61.04$, $SD = 9.95$), as in $t(118) = 1.08$, $p = .283$, $d = 0.197$, thus, H1a is rejected.

As explained in section 4.1, performance can be described according to 2×2 a version of the grid used in SDT to allow bias towards accepting Stylus advice to be separated from ability to distinguish correct sentences. Table 2 shows the SDT grid populated with the number of responses and the response rates of the current experiment.

Table 2. Number of responses and response rate per category per group.

	Stylus suggestion correct	Original sentence correct
GGood		
Stylus suggestion selected by participant	$N_H M = 13.34 (SD = 4.12)$ $H = 13.34/20 = 0.67$	$N_{FA} M = 3.85 (SD = 1.86)$ $FA = 3.85/10 = 0.39$
Original sentence selected by participant	$N_M M = 6.66 (SD = 4.12)$ $M = 6.66/20 = 0.33$	$N_{CR} M = 6.15 (SD = 1.86)$ $CR = 6.15/10 = 0.61$
GBad		
Stylus suggestion selected by participant	$N_H M = 5.79 (SD = 1.46)$ $H = 5.79 /10 = 0.58$	$N_{FA} M = 7.48 (SD = 2.66)$ $FA = 7.48/20 = 0.37$
Original sentence selected by participant	$N_M M = 4.21 (SD = 1.46)$ $M = 4.21 /10 = 0.42$	$N_{CR} M = 12.52 (SD = 2.66)$ $CR = 12.52 /20 = 0.63$

5.2. Testing sensitivity and bias

Mean d' was higher in GGood ($M = 0.85$, $SD = 0.98$) than in GBad ($M = 0.55$, $SD = 0.53$), but as Levene's test was significant ($p < .05$), which indicates lack of equal variance, no t -test was performed.

H1b Compared to GBad participants, participants in GGood will during the task have a stronger bias towards following Stylus' advice, measured using Signal Detection Theory.

Mean c was significantly higher in GGood ($M = 0.18$, $SD = 0.76$) than in GBad ($M = -0.12$, $SD = 0.56$), as in $t(118) = 2.52$, $p = .013$, $d = 0.460$. Hence, H1b is confirmed, which confirms our prediction that participants in GBad notice Stylus' poor performance and therefore feel less inclined to accept its advice. Neither groups' bias is very strong though; if they would have followed the aid blindly, c would be 2.33, it would have been -2.33 if they would always have gone against the advice, and 0 had they completely ignored it.

5.3. Testing the role of pre-task trust and perceived self-efficacy

The bias scores also allow us to test the main hypotheses concerning the role of prior trust and self-efficacy in determining the propensity to accept advice.

H2a Independent of the correctness of its advice, participants will have a higher propensity to accept the aid's suggestions if their prior trust in automated writing style checkers is high.

The acceptance of Stylus recommendations is the degree of bias participants displayed. In GGood there was a significant correlation between prior trust in writing style checkers and c , $r(57) = -.27$, $p = .042$. This shows that the higher GGood participants' level of trust in similar systems is, the more likely they are to accept the aid's advice, and vice versa, as observed earlier by others in other domains. Likewise in GBad, there was a significant correlation between prior trust in writing style checkers and c in GBad, $r(59) = .26$, $p = .042$.

H2b Independent of the correctness of its advice, participants will have a higher propensity to accept the aid's suggestions if their perceived self-efficacy in the domain of writing is low.

In GGood there was no significant correlation between perceived self-efficacy and bias toward accepting Stylus suggestions, c , $r(57) = -.04$, $p = .771$, and there was no significant correlation either in GBad, $r(59) = -.04$, $p = .776$. Thus, H2b is rejected.

Additionally, the sensitivity scores allow us to check whether participants' prior perceived self-efficacy, which although as already shown is likely overestimated, but as suggested in the literature can reliably predict their level of performance. In GGood, there was no significant correlation between participants' prior perceived self-efficacy and their sensitivity d' , $r(57) = .14$, $p = .309$, nor did we find a significant correlation in GBad, $r(59) = -.01$, $p = .917$.

5.4. Confidence analysis

The average self-reported confidence across the task was 89.45 ($SD = 7.15$) for GGood, and $M = 86.70$ ($SD = 8.11$) for GBad. The difference in confidence between the groups is tested in the ANOVA later in this section. Table 3 shows a breakdown the confidence scores for both groups. In GGood, 2 participants had $N_M = 0$, hence no M confidence ratings, and 2 participants had $N_{FA} = 0$, hence no FA confidence ratings. In GBad, 2 participants had $N_M = 0$, hence no M confidence ratings. To substitute a missing confidence value, the mean of a participant's own present values was used.

The confidence data in Table 3 were analysed in a $2 \times 2 \times 2$ mixed ANOVA, with Group (GGood vs. GBad) as a between-subjects factor, and Correctness of response (Correct vs. Incorrect) and Type of response (Stylus vs. Original) as within-subjects factors. The correctness-of-response label has been added to the cells in Table 3 to clarify this retabulation, which allows Correctness of response to be treated as a main effect.

Table 3. Mean confidence percentage per category per group.

	Stylus suggestion correct	Original sentence correct
GGood		
Stylus suggestion selected by participant	Response correct $M = 90.16$ ($SD = 7.70$)	Response incorrect $M = 89.58$ ($SD = 8.18$)
Original sentence selected by participant	Response incorrect $M = 84.87$ ($SD = 10.25$)	Response correct $M = 90.58$ ($SD = 7.73$)
GBad		
Stylus suggestion selected by participant	Response correct $M = 87.39$ ($SD = 8.38$)	Response incorrect $M = 85.01$ ($SD = 9.50$)
Original sentence selected by participant	Response incorrect $M = 85.87$ ($SD = 10.01$)	Response correct $M = 87.50$ ($SD = 8.80$)

The ANOVA showed a significant main effect of Correctness of response, in that correct responses ($M = 88.89$, $SD = 7.62$) were assigned more confidence than incorrect responses ($M = 86.29$, $SD = 8.67$), $F(1, 118) = 30.166$, $p < .001$, *partial* $\eta^2 = 0.204$. There was no significant main effect for Type of response, i.e. Stylus ($M = 88.01$, $SD = 8.03$) vs. Original ($M = 87.17$, $SD = 8.47$), $F(1, 118) = 3.134$, $p = .079$, *partial* $\eta^2 = 0.026$. There was no significant between-subjects main effect of Group, $F(1, 118) = 2.711$, $p = .102$, *partial* $\eta^2 = 0.022$.

The repeated measures ANOVA also revealed two significant two-way interaction effects: The first is for Type of response x Group, $F(1, 118) = 7.60$, $p = .007$, *partial* $\eta^2 = 0.06$. The second is Correctness of response x Type of response, $F(1, 118) = 7.436$, $p = .007$, *partial* $\eta^2 = 0.059$. There was no significant two-way interaction of Correctness of response x Group, $F(1, 118) = 1.603$, $p = .208$, *partial* $\eta^2 = 0.013$. Lastly, there was a significant three-way interaction, Correctness of response x Type of response x Group, $F(1, 118) = 13.251$, $p < .001$, *partial* $\eta^2 = 0.101$.

Inspecting the means in Table 3, we can see how all these interaction effects reflect the fact that only in GGood, incorrect Stylus responses are made more confidently than incorrect Original responses. The mean confidence of 84.87 for GGood incorrect Original responses appears to be anomalously low. It is possible to speculate about reasons for this effect (it is unsurprising that confidence might be low when the participant doesn't know the right answer and is going against the advice of a rather reliable aid). However, we do not feel this is a clear enough effect to warrant further discussion without more data.

H3 Participants in both groups will be overconfident, especially when confidence is measured trial-by-trial; i.e. their estimates of percentage correct will be higher than actual achieved percentage correct.

The average trial confidence was significantly higher than performance in GGood, as in $t(58) = 10.54$, $p < .001$, $d = 0.259$, and in GBad, as in $t(58) = 16.18$, $p < .001$, $d = 2.072$, hence H3 is confirmed.

5.5. Post-task measures

GGood's subjective estimation of the number of times they selected the correct answer (either Original or Stylus), $M = 21.05$ ($SD = 5.85$), was higher than their actual number of correct responses, $M = 19.49$ ($SD = 4.87$). Across participants, estimated performance did not correlate significantly with objective performance, $r(57) = .04$, $p = .74$. GBad's average estimation was 21.43 ($SD = 5.32$), which was again higher than actual performance, $M = 18.31$ ($SD = 2.99$). Again, there was no significant correlation between estimated and actual performance across participants in the group, $r(59) = -.16$, $p = .219$. Interestingly, participants' online observation of the correctness of their responses was more accurate, in that confidence in correct responses was higher than in incorrect ones, as described in section 5.4.

We also compared the number of times participants thought they chose the Stylus suggestion over the Original sentence with the objective frequencies. In GGood, the average number of times participants thought they chose the Stylus suggestion was 17.53 ($SD = 5.59$), which is slightly higher than their actual number of Stylus choices, $M = 17.19$ ($SD = 4.15$). However, across participants the estimated number of Stylus choices did not correlate significantly with the objective number, $r(57) = .03$, $p = .836$. In GBad, the group that encountered the poorer performing version of Stylus, the average estimated number of Stylus choices was 15.23 ($SD = 5.26$), which is higher than the observed number, $M = 13.26$ ($SD = 3.08$). Again, there was no significant correlation between the estimated number of Stylus responses and the actual number in this group either, $r(59) = .10$, $p = .457$.

H4 Compared to GBad participants, participants in GGood will develop more trust in Stylus during the experiment, as measured post task.

As predicted, post-task trust in Stylus was significantly higher in GGood ($M = 67.97$, $SD = 17.07$) than in GBad ($M = 58.22$, $SD = 17.20$), as in $t(118) = 3.10$, $p = .002$, $d = 0.57$. Thus, H4 is confirmed, which shows the effect of Stylus' reliability on participants' post task trust, and confirms that users recognise the aid's reliability.

6. Conclusions and discussion

In this article we set out to contribute to better understanding complex knowledge-based cognitive tasks of interaction with imperfect automated aids in the domain of spelling and grammar checking, by developing a novel experimental paradigm based on SDT. In an experimental study where participants had to judge sentences with assistance from a purported automated aid, we tested hypotheses around effects of performance, trust, and confidence. The use of the SDT-construct of bias, as a measure of users' propensity to accept suggestions from an automated aid, is one of the major novel methodological contributions.

It has been argued in the Human Factors literature that one plausible model for the use of automated assistance in decision-making is that it is influenced by the interplay of users' trust in comparable automated aids and their perceived self-efficacy (i.e. trust in their own ability), and that this in return affects the confidence they have in their own ability (Lee and Moray 1994; Moray et al. 1994; Riley 1989, 2018; Wiczorek and Meyer 2019) and the trust they have in an aid. This very simple model of trust and confidence operating in balance has some support in the area of perceptual decision-making and simple control tasks, but it has hardly been explored at all in the domain of knowledge-dependent cognitive skills of the kind for which automated assistance has become more prevalent. Thus, the first contribution of this study is to begin an exploration of personal beliefs in relation to performance under uncertainty (Edwards 1954), and with support from an imperfect automated aid (Wickens and Dixon 2007) in the domain of text writing and editing, and in particular spelling and grammar checking. Not only are writing and editing aids a common current example of automated assistance that perhaps represent aspects of the use of similar aids in other domains, but this application also allows for a great level of experimental flexibility and efficiency at reasonable cost.

6.1. Summary and discussion of key findings

6.1.1. Sensitivity and bias

We statistically compared sensitivity and bias of two socio-demographically equal groups who performed the same task, but were assisted by a purported automated aid that performed at a different level of reliability for each group.

By analysing bias, we could compare participants' preference for the aid's suggestions independent of their correctness. There was a significant difference in bias between the groups, which confirms that participants in

GGood recognised the fact that correct Stylus responses were more prevalent than Original ones, and vice versa in GBad. Scores were close to 0 in both groups, which implies, unsurprisingly, that participants' own judgement of sentences was the primary step, with the aid's suggestion serving as a secondary line of advice. It seems likely that the rather weak performance of the aid in both conditions contributes to this finding.

The violation of equal variance in some of the data dictates that we treat above test results with caution, and interpret them as suggestions or patterns that will be further researched in future studies, rather than as statistically significant results.

6.1.2. Effects of trust and perceived self-efficacy

In both groups there was a significant correlation between participants' prior trust in writing style checkers and their bias towards the aid. We believe the result provides some confirmation of the predicted pattern in our data. On the other hand, we found no evidence at all that perceived self-efficacy affected bias, which means that only one half of the proposed balance between trust and self-efficacy in the use of automated aids that is suggested in the literature received any support.

6.1.3. Confidence

If confidence is well-calibrated, participants' confidence in their correct responses should be higher than their confidence in incorrect responses; this was confirmed in our findings. We also found a significant difference in participants' average confidence from the trials between the groups, with GGood being overall more confident than GBad. There were three significant confidence interaction effects as well, Type of response x Group (i.e. Stylus vs. Original in each group), Correctness of response x Type of response (i.e. H, M, CR and FA), and Correctness of response x Type of response x Group (i.e. proportion of H, M, CR and FA between the groups).

One overconfidence measure is participants' average confidence in their responses in each trial, which can be compared with their percentage of correct responses. Although participants' estimation of performance was on average higher than their actual percentage correct, we did not find a significant correlation between the two.

6.1.4. Post-task trust

We assumed that participants who received suggestions from a better performing system would develop more trust in the system than participants who worked with a system that performed worse. We found that GGood's trust in the aid was indeed significantly higher than GBad's.

7. Concluding remarks

We demonstrated that the influence of trust in similar systems has a smaller effect on users' performance during a task than anticipated on basis of the literature (Bisantz and Seong 2001; Chavaillaz, Wastell, and Sauer 2016; Dzindolet et al. 2003; Lee and Moray 1992; Moray et al. 1994; Moray and Inagaki 1999; Muir 1987). We found no evidence that perceived self-efficacy had a noticeable influence on performance. Users interacting with the most reliable version of the aid had the most confidence in their own responses.

Another characteristic that this experiment confirms, is that users can recognise how well an aid is doing, even if they do not receive any feedback on its performance (Hutchinson et al. 2022). Users of a better performing system showed to be more willing to accept the aid's advice, which shows an effect of the *reliability* of the automation's suggestions. Without receiving feedback about the correctness of their responses, users also show they have a sense of their own performance, which is demonstrated by a higher level of confidence in correct responses than in incorrect ones.

Our research has demonstrated opportunities and limitations of using SDT to analyse effects of performance, trust, and confidence in cognitive tasks aided by imperfect automated aids, and of researching participants' propensity to accept automated advice as 'bias'. This approach overcomes a problem with standard methods for measuring reliance on an aid, which depend on an independent recording of unaided biases too. Our research will be valuable for future research in the domain of automatically aided text writing and editing as well as in other domains, especially in combination with other research that explores individual or adjustable alerting thresholds (see e.g. Meyer and Sheridan (2017) and Gadala, Strigini, and Ayton (2021)).

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